ThermalScout-Net: Design and Evaluation of a Custom Deep-Learning Detector for Nighttime Wildlife Surveillance

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***Abstract*—Nighttime thermal drone imagery can support wildlife monitoring, but the small size of distant people and animals makes automatic detection difficult. This project proposes *ThermalScout-Net*, a student-designed, lightweight convolutional object detector that locates and labels humans and animals in thermal video frames. The proposed network will combine a feature-extraction backbone with a custom multi-scale feature-fusion module and detection heads, then compare randomly initialized and pretrained versions against a YOLO11n transfer-learning baseline. Research-driven model selection and a small randomized hyperparameter search will investigate the trade-off among detection accuracy, human recall, inference speed, and computational cost. Experiments will use the real-video portion of the BIRDSAI dataset with a video-disjoint test split. This document presents the proposed design and evaluation plan; no detection results are claimed.**

***Index Terms*—thermal infrared, unmanned aerial vehicles, object detection, wildlife conservation, convolutional neural networks, transfer learning, model selection**

I. Introduction and Problem Statement

Thermal infrared (TIR) cameras can observe people and wildlife at night when ordinary visible-light imagery is limited. However, targets may occupy only a few pixels, while moving cameras, reflections, clutter, and occlusion produce false detections. Bondi *et al.* introduced BIRDSAI specifically to study detection and tracking in aerial TIR footage from conservation settings [1]. In an anti-poaching support scenario, an algorithm should highlight potential people and animals for a human analyst; a detected person must *not* be treated as a confirmed poacher.

The project problem is: *Can a compact, student-designed object-detection network identify and localize humans and animals in unseen thermal drone videos, and how does its performance compare with a modern pretrained detector?* An image classifier alone is insufficient because it does not report the location of each object. The proposed system will produce bounding boxes, class labels (human or animal), and confidence scores for each frame.

II. Research Basis and Project Contribution

A. Selecting the Detection Approach

Two-stage methods such as Faster R-CNN explicitly generate candidate object regions and then classify or refine them [4]. Single-stage detectors such as YOLO directly predict object locations and classes, making them useful baselines when efficient inference matters [5]. Their relative performance on BIRDSAI must be *measured*, not assumed. For a time-limited course project, YOLO11n will be the principal standardized baseline; Faster R-CNN will be reviewed as a two-stage research alternative and trained only if resources permit.

The proposed backbone uses the efficient MobileNetV3-Small family [3], with a *newly implemented* thermal-image adaptation, feature-fusion stage, and objectness/class/box prediction heads. This is an original *application-specific detector implementation and experimental design*, not a claim to have invented an entirely new neural-network paradigm. Two versions of the same custom architecture will separate architecture effects from pretraining effects.

B. What "Search" Means in This Project

Here, searching for the best model means *empirical architecture selection and hyperparameter optimization*, not applying BFS or A* to individual pixels. The study will first compare candidate detection approaches using published research, then conduct a limited randomized search over training settings. Random search is a defensible baseline for hyperparameter tuning under fixed budgets [6]. Model selection will use validation data only.

III. Technical Objectives

The project's measurable objectives are to: (1) implement a custom PyTorch detector that outputs multiple bounding boxes per frame; (2) train the same architecture from random initialization and from pretrained backbone weights; (3) train and benchmark a YOLO11n transfer-learning model; (4) measure human and animal detection independently, especially for small targets; and (5) document accuracy–speed trade-offs and failure cases.

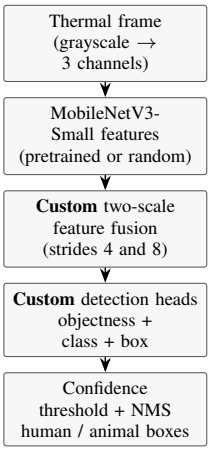


Fig. 1. Proposed ThermalScout-Net processing pipeline; custom modules are highlighted in bold.

IV. Proposed System Design

A. Dataset and Preparation

Experiments will use BIRDSAI's real nighttime thermal videos, released through Conservation Drones [1], [2]. The collection includes 48 real videos, with an official partition of 32 training and 16 test videos in the paper. The dataset annotations use the MOT-style fields `frame, object_id, x, y, w, h, class, ...`, where class 0 denotes animal and class 1 denotes human [2]. A validation set will be created by *entire video*, exclusively from the original training videos; test videos will remain untouched until final evaluation.

An annotation converter will match labels to frame names, validate box coordinates, remove invalid boxes, and export the data to a consistent two-class detection format. Grayscale thermal frames will be normalized and, for compatibility with ImageNet-pretrained backbones, replicated across three input channels. Aspect ratios and bounding-box coordinates will be preserved when resizing. Sampling will be performed by video rather than by arbitrary adjacent frames, limiting near-duplicate leakage.

B. Student-Designed ThermalScout-Net

The network will use MobileNetV3-Small convolutional features at approximately 1/4 and 1/8 image resolution. Custom lateral layers will fuse fine spatial detail with richer features; lightweight anchor-free detection heads will predict (i) objectness, (ii) two class scores, and (iii) four bounding-box distances at candidate locations. This higher-resolution branch is motivated by small thermal objects. During inference, confidence filtering and non-maximum suppression (NMS) will reduce duplicate boxes. Label assignment, box decoding, and loss computation will be written and tested explicitly rather than delegated to a prebuilt YOLO training loop.

For training, a weighted multi-task objective will combine objectness, classification, and box regression:

$$\mathcal{L} = \lambda_o \mathcal{L}_{\text{focal}} + \lambda_c \mathcal{L}_{\text{cls}} + \lambda_b \mathcal{L}_{\text{box}}. \tag{1}$$

TABLE I
Planned Detector Comparisons

ID	Detector	Initial weights
A	ThermalScout-Net (custom)	Random
B	ThermalScout-Net (custom)	MobileNetV3 pretrained
C	YOLO11n baseline	COCO pretrained
D*	YOLO11n baseline	Random

*Stretch goal; not required for the minimum comparison.

Here, focal loss addresses the abundance of background positions, classification loss separates humans from animals, and a box-regression loss (e.g., Smooth L1) learns target locations. The weights λ_o, λ_c, and λ_b will be fixed initially and tuned using validation performance if needed.

V. Experimental Plan

A. Controlled Comparisons

Table I identifies the planned minimum experiments. Each experiment will use the same real-video partition, object classes, evaluation script, and documented image scale. Training duration, hardware, and any differences in augmentation will be reported. A randomly initialized YOLO11n is optional if time or GPU capacity allows.

B. Architecture/Hyperparameter Search

After an initial baseline run, a small, reproducible randomized search (approximately four to six configurations) will vary learning rate, input resolution, and detection threshold or loss balance. The number of trials will be capped to fit available compute, and the same validation criterion will rank candidates. Resolution and augmentation ablations will test whether small-object detection benefits from greater spatial detail. The test set will *not* be used to select a model or threshold.

C. Evaluation and Analysis

Primary detection metrics will be mean average precision at intersection-over-union (IoU) 0.50 (mAP@0.50) and mAP averaged over IoU thresholds 0.50–0.95. Precision, recall, and F1 will be reported for each class, together with confusion/error examples. Special emphasis will be placed on human recall, because missing a small person can be operationally important; its false-alarm cost will also be reported. Inference latency (ms/frame), frames per second on named hardware, model parameters, and training time will quantify efficiency. Visual results will include correct detections, missed objects, and false positives grouped by target size and scene characteristics. No performance target is presented as an achieved result.

VI. Feasibility, Risk, and Responsible Use

The minimum deliverable consists of the dataset converter, working custom detector, scratch-versus-transfer comparison, YOLO11n baseline, evaluation tables, plots, and annotated video frames. CPU-only training may require a modest sampled subset and fewer epochs; GPU execution or Colab will

be used if available. Risks include imbalanced human/animal counts, extremely small targets, incorrect frame-to-label mappings, overfitting adjacent video frames, and training instability. Mitigations include file/box unit tests, visual annotation audits, stratified reporting, early stopping, and video-level splits. This is a research prototype for *human-reviewed* surveillance, not a real-time field deployment or an automatic judgment of suspicious behavior. Privacy, misidentification, and domain shift to other cameras or environments will be discussed.

VII. Schedule and Expected Deliverables

Phase 1: verify real-video annotations, complete background research, choose the comparison protocols. **Phase 2:** implement and debug ThermalScout-Net; confirm that a small sample can be overfit as a sanity test. **Phase 3:** run scratch, transfer-learning, and YOLO11n experiments; tune on the validation videos. **Phase 4:** test the chosen checkpoints once, analyze failure cases, and prepare the final IEEE-style report and presentation. Additional two-stage benchmarks or temporal tracking are stretch goals only.

VIII. Expected Contribution

The core contribution will be an auditable custom thermal object detector and a controlled study of when transfer learning helps, how architecture choice changes speed and accuracy, and why tiny thermal targets remain difficult. Regardless of which detector wins, the project will demonstrate AI model design, informed experimental selection, and critical evaluation rather than only running a pretrained model.

References

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